

Conclusions: We developed a multi-channel infrared-fMRI system and demonstrated its feasibility of producing mesoscale stimulus with varying channel spacing and mapping static networks of two separated points. The overall network with separable channel-specific clusters suggests the possible coordinated columnar structure across the whole brain. And the network variability across the hierarchy indicates that, similar to the structural architecture, a regularity of functional connectivity at the mesoscale might exist.

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Poster No 38

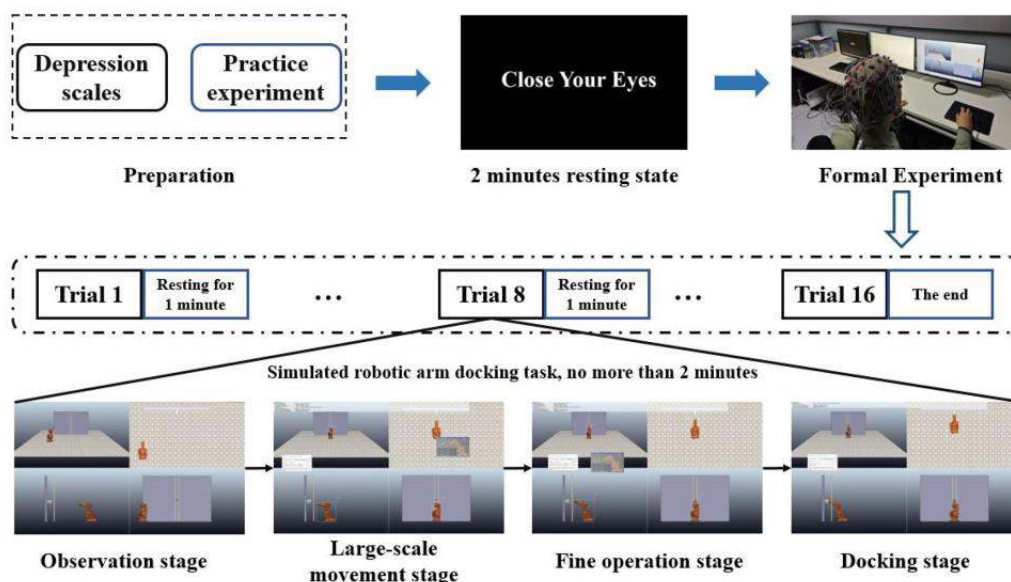
Subthreshold Depression Affects EEG Network Patterns in Spatial Cognition Process

Kai Yang¹, Yidong Hu¹, Ying Zeng¹, Li Tong¹, Bin Yan¹

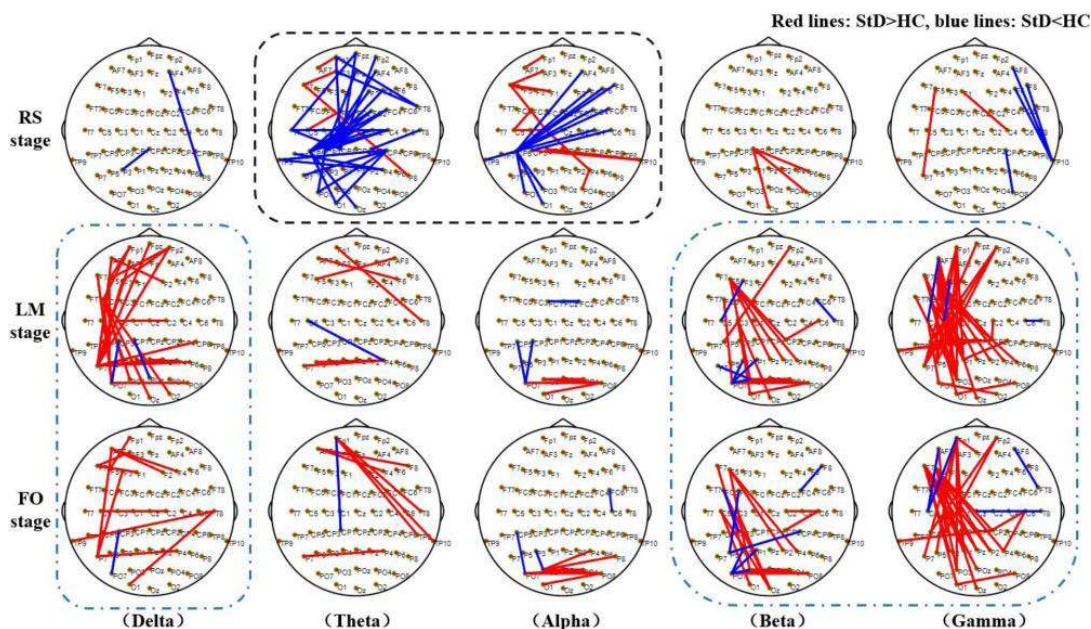
¹Strategic Support Force Information Engineering University, Zheng Zhou, He Nan, China

Introduction: Subthreshold depression (StD) refers to clinically relevant depressive symptoms that do not meet the criteria for depression (Hwang et al. 2016). Many studies have reported cognitive impairments in individuals with StD. Especially, spatial cognition impairment can have a severe impact on the StD individuals' ability to judge distance and orientation in daily life (Volz et al. 2022). However, the patterns of neural activity during spatial cognition processes in individuals with StD remain unclear. To explore the impact patterns of StD on spatial cognition, our study designed a spatial cognition experiment that mimics a natural scene and analyzed the EEG networks during both resting and task states.

Methods: According to the definition of StD in previous studies (Volz et al. 2022), this work screened 15 StD and 15 healthy control (HC) subjects each through interviews, the Center for Epidemiological Studies Depression Scale (CES-D, Chinese version) and Beck Depression Inventory-II (BDI-II) scale. The experiment included a 2 minutes eyes-closed resting state (RS) stage and 16 simulated robotic arm docking tasks. The docking tasks were close to natural scenes to induce the spatial cognition of subjects. The EEG data were recorded by 62 channels G.Hlamp system and the operation details were recorded through screen recording software. Then each docking process was divided into four stages: observation stage, large-scale movement (LM) stage, fine operation (FO) stage and docking stage, the operation time of each stage were got by offline analysis of screen recording videos. The main steps of EEG preprocessing include: data segmentation, average reference, 1-60Hz filtering, artifact removal, baseline correction, trial rejection (>100μV). Then the preprocessed EEG data were filtered into five frequency bands: delta (1-4Hz), theta (4-8Hz), alpha (8-13Hz), beta (13-30Hz), and gamma (30-50Hz). In each frequency band, the phase locking values of every two EEG channels were calculated, and there were five 62*62 network matrixes. At the end, networks of StD group and HC group were compared by t-test.



Results: The results of operation time showed that StD group spent significantly more time than HC group ($P < 0.05$) in LM stage and FO stage while completing the docking task. The longer operation time of the StD group validated the spatial cognition impairment of StD subjects in previous studies. Then, this study focused on analyzing brain networks of RS stage, LM stage and FO stage. For the RS stage, in the theta and alpha bands, the HC group had stronger connections with the channels in parietal lobe as hub nodes compared to the StD group. Parietal lobe was related to spatial attention, the differential connections in RS stage may indicate that the StD caused spatial attention decline (Aleksandra et al. 2023). For LM and FO task stages, in the delta, beta and gamma bands, the frontoparietal connections of StD group were stronger than those of HC group. These enhanced frontoparietal connections may be neural compensation made by the StD group to complete spatial cognition task (Vallesi et al. 2011; Caldinelli et al. 2022). In the whole, network patterns of RS stage indicated the spatial cognition decline of StD group, and the enhanced frontoparietal connections in LM and FO task stages, possibly indicated the neural compensation mechanism of StD group to ensure spatial cognition task completion.



Conclusions: This study analyzed the impacts of StD on spatial cognition based on resting and task state EEG networks. Consistent with previous studies, behavioral results and RS network patterns indicated that StD can lead to spatial cognition decline. At the same time, there was neural compensation mechanism of StD group by enhancing frontoparietal connections to ensure the completion of spatial cognition tasks. This study provides new insights into the impact of StD on spatial cognition.